



PHD & POSTDOCTORAL RESEARCH

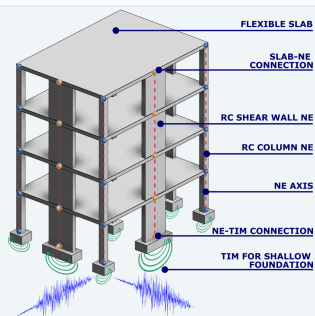
Better models. Better decisions.

Join our work in earthquake engineering, computational mechanics, AI-driven simulation and early-age concrete behavior at Universidad de los Andes.

TWO RESEARCH PROJECTS · ONE COLLABORATIVE GROUP

Research with a purpose.

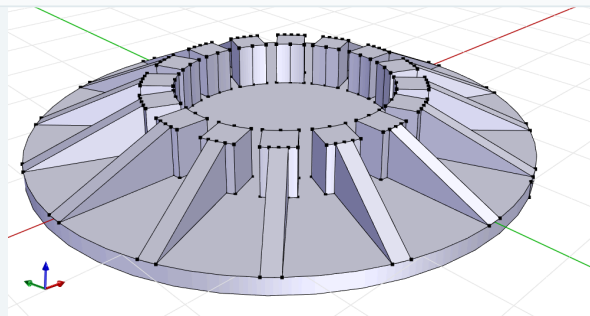
We connect physics-based simulations, experiments and open-source tools to questions engineers actually face. Choose a direction and help shape it.



FONDECYT REGULAR 1261870

Seismic design, AI & HPC

Develop neural structural elements and efficient soil–foundation models trained on high-fidelity simulation. Explore and validate performance-based designs with OpenSees and HPC.



FONDEF IDEA I+D ID26110834

Prevent concrete cracking

Combine thermo-mechanical simulation, laboratory experiments and on-site monitoring to predict and mitigate early-age cracking in massive concrete elements.



Find your path.

A doctoral research route and a postdoctoral route, each connected to the group's current projects.

01 · DOCTORAL RESEARCH

PhD candidates

Chilean and international applicants welcome. Enroll in the Doctorado en Ciencias de la Ingeniería at Universidad de los Andes.

An ANID doctoral scholarship award is required for this PhD opportunity. I can support you in developing the research proposal and application.

02 · POSTDOCTORAL RESEARCH

Postdoctoral researchers

Chilean nationals only under the ANID provisions applicable to these opportunities.

Bring a research direction that connects with one of the projects. Contact me to discuss fit and the relevant funding arrangements.

The PhD opening requires an ANID scholarship award. Postdoctoral eligibility is restricted to Chilean nationals under ANID provisions; confirm funding details before applying.

What you'll work with

- OpenSees, ShakerMaker and open-source research tools
- Nonlinear finite elements, seismic wavefields and scientific ML
- Multi-node HPC, laboratory testing and field observations

What helps you thrive

Structural or geotechnical engineering, concrete thermo-mechanics, scientific programming, machine learning or instrumentation. No one needs every skill—bring curiosity and a willingness to learn across disciplines.

Start a conversation.

Email a short CV, the project and route you are interested in, and a few lines about the questions you want to tackle. PhD candidates: tell me where you are in the ANID scholarship process.

jaabell@uandes.cl · [Read the full opportunity ↗](#)